Calculus of Variation and Tensor Calculus

Final paper: Applications of Jacobi Field in Geodesic Flow

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December 12, 2023

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0 Introduction

Mathematics abstracts the structures from reality, and abstracts structures from structures. Throughout this paper, we will see many concepts based on:

"set"(The underlying objects) + "structures"(The properties/theory)

By piling up structures layer by layer, this paper explain some ideas of differential geometry, and give a brief introduction to the dynamical properties of geodesic flow. As a formal mathematical formulation of affine connections and geodesics, section 1 is retained in order to make the paper self-contained.

1 Preliminaries

In this part we review some important facts of smooth manifolds, tensors and variational calculus.

We start from manifold without any additional structure: "topological manifold", then we put the smooth structure on it, to make it "smooth(differentiable) manifold", we then put the Riemannian structure on it to get "Riemannian manifold", and finally get the induced "affine connection(parallel transport) structure". Riemannian structure and affine connection structure are independent, we can define different affine connections on smooth manifold without Riemannian metric. However, under the metric compatibility and torsion-free assumption, we can uniquely determine a affine connection on this smooth manifold.

1.1 Topological Manifolds

In Euclidean space, everything is flat and idealized, therefore it might not be strong enough for us to conceptualize some ideas. Manifold is somewhere we can generalized the idea of continuous function in Euclidean space to a curved space. This is a space locally modeled by Euclidean space.

The mathematical starting point of continuity is topology. A **manifold** is a "set" with "topological structures": Hausdorff, second countable, and locally homeomorphic to a open set in $\mathbb{R}^n$, we usually call this open set a **chart**. In most cases we are not considering the crazy manifolds, so the Hausdorff and second countability can sometimes take for granted. The locally Euclidean condition is that we can use the locally rectilinear coordinate to represent the curvilinear coordinate. According to the big theorem of Whitney, we can always think the manifold of some subset of $\mathbb{R}^{2n+1}$, where n is the dimension of our manifold.

1.2 Smooth Manifolds

Topological manifold is not strong enough to do calculus on it. We need smooth structure, which forms a smooth manifold. What is different from topological manifold is that we need a local diffeomorphism to Euclidean space, stronger than homeomorphism. This is because we want the smoothness of local representation not to be affected by coordinate transformation from one chart to another.

Tangent Space The tangent space of M at p, is denoted as T_pM. We naturally think of tangent space as a linear approximation around a point on a manifold. A concrete way to think about it is we can take the tangent space as the equivalent classes of same-direction vectors, and of course, locally. As we will see, those tangent vectors can be identified as derivations, namely, $T_pM \cong \mathbb{R}^n$.

Derivation A derivation is a $\mathbb{R}$-linear operator $\omega_p : C^\infty(M, \mathbb{R}) \to \mathbb{R}$ satisfying:

1. $\omega_p(fg) = f(p)\omega_p(g) + g(p)\omega_p(f)$
2. $\omega_p(a_1 f_1 + a_2 f_2) = a_1\omega_p(f_1) + a_2\omega_p(f_2)$
3. $(c\omega_p)(f) = c(\omega_p(f))$

Another definition of the tangent space of M at p, denoted $T_pM :=\{\text{Derivations of } C^\infty(M) \text{ at } p\}$. T_pM is a real vector space.

Take $U \subset \mathbb{R}^n$ as the manifold. We shall have a one-to-one correspondence between $\mathbb{R}^n$ and T_pM, namely,

$$\begin{aligned} \varphi : \mathbb{R}^n &\to T_p\mathbb{R}^n \\ e_i &\mapsto \left.\frac{\partial}{\partial e_i}\right|_p \end{aligned}$$

Where e_i is a standard basis of $\mathbb{R}^n$.

Reversely, if we consider the Taylor expansion of a smooth function, we can construct the inverse image.

Hence, $\{\frac{\partial}{\partial e_1}, \frac{\partial}{\partial e_2}, ..., \frac{\partial}{\partial e_n}\}$ becomes a basis of $T_p\mathbb{R}^n$, so all the derivation can be written as a linear combination of this basis.

Tangent Bundle Tangent bundle is defined as

$$\bigsqcup_{p \in M} (p, T_pM)$$

This is a $2n$-dimensional smooth manifold as long as M is a n-dimensional smooth manifold. We can do calculus on it. However, this is not meaning that $TM \simeq M \times \mathbb{R}^n$. The tangent bundle is not a trivial bundle in general.

Differential as Linear Mapping In calculus, we take the derivative as the best linear approximation. Similarly, we interpret the differential of a function $f : M \to N$ as a linear mapping between two tangent spaces, namely: $Df : T_pM \to T_{f(p)}M$, which is defined as follows:

$$\begin{array}{rccc} v \mapsto f_*(v) : & C^\infty(M) & \to & C^\infty(M) \\ & g & \mapsto & v(g(f)) \end{array}$$

Smooth Vector Field A smooth vector field X on a manifold M is a smooth map that assigns to every point $p \in M$ a vector X_p in the tangent space T_pM at p (So this is globally defined rather than locally). We give every point p on the manifold M a vector in T_pM, then it is a vector field.
Every tangent vector can be uniquely represented as a linear combination of basis of this tangent space, we can always write the vector field as

$$X = \sum_{i=1}^{m} v_i \frac{\partial}{\partial x_i}$$

Where $v_i : U_\alpha \to \mathbb{R}$ are smooth real-valued functions for all i.

Partition of Unity Given a smooth manifold M, and $\{\mathcal{C}_\alpha\}_{\alpha \in I}$ an open cover of M. There are countably many compactly supported $\lambda_i \in C_c^\infty(M, \mathbb{R})$ such that:

1. For all i, $\text{supp}\lambda_i \subset C_\alpha$ for some α.
2. $\{\text{supp}\lambda_i\}$ is locally finite, namely, for any $p \in M$, exists an open neighbourhood V of p, only finite sets in $\{\text{supp}\lambda_i\}$ has nonempty intersection with V.
3. $\lambda_i \geq 0$, and $\sum_{i=1}^{\infty} \lambda_i \equiv 1$

This theorem is mainly saying that we can smoothly partition the function on a manifold. We sometimes cannot write a function on manifold globally, but by partition of unity, we can consider functions in the local neighbourhood and glue them together finally. The function,

$$f(t) = e^{-\frac{1}{t}} u(t)$$

where $u(t)$ is Heaviside function, is a smooth function. This gives a smooth way to connect 0 and 1. Actually, the proof of partition of unity is based on this idea.

1.3 Riemannian Manifolds

On smooth manifold, we can do calculus. However, when talking about the distance, smooth manifold do not have enough structures. Then comes the Riemannian metric tensor.

Riemannian Metric We want to define distance on the manifold, therefore we need to generalize the Euclidean inner product. The smooth manifold M, together with a Riemannian metric g, denotes (M, g), is called a Riemannian manifold.
We write the metric tensor as

$$ds^2 = \sum_{i,j=1}^{m} g_{ij} dx^i \otimes dx^j$$

it is an inner product, a bilinear form with the (i, j)-th entry $g_{ij} = \langle e_i, e_j \rangle$. We sometime write it as $ds^2 = g_{ij} dx^i dx^j$ to emphasis the symmetric property of g_{ij}.

We therefore can define the length of a curve on M. We first parameterize it:

$$\gamma : [0, 1] \to M$$

$$L(\gamma) = \int_0^1 \sqrt{\langle \dot{\gamma}, \dot{\gamma} \rangle} dt = \int_0^1 \sqrt{g_{ij} \frac{d\gamma^i}{dt} \frac{d\gamma^j}{dt}} dt$$

This makes the Riemannian manifold a metric space. We know a connected Riemannian manifold is also path connected, so we can have $L(a, b) = \inf_\gamma L(\gamma)$, where γ is a curve on M connecting $a, b \in M$.

Pseudo-Riemannian Metric A Riemannian metric requires a positive definite metric, but from linear algebra we know that quadratic forms may not necessarily be positive definite. We can generalize the notion of a Riemannian manifold to something called a pseudo-Riemannian manifold, which is a manifold M with a metric g which may not be positive definite; for example, maybe the metric is of the form $g = dx_1^2 + ... + dx_p^2 - dx_{p+1}^2 - ... - dx_{p+q}^2$.

1.4 Affine Connection

Definition of Affine Connection Recall the directional derivative of function $f \in C^\infty, \mathbb{R}^n \to \mathbb{R}^m$ along $\mathbf{v}$, we have:

$$D_{\mathbf{v}} \mathbf{f}(\mathbf{x}) = \lim_{h \to 0} \frac{\mathbf{f}(\mathbf{x} + h\mathbf{v})}{h}$$

It has the following properties:

1. $D_{\alpha \mathbf{v} + \beta \mathbf{w}} \mathbf{f} = \alpha D_{\mathbf{v}} \mathbf{f} + \beta D_{\mathbf{w}} \mathbf{f}$
2. $D_{\mathbf{v}}(\alpha \mathbf{f} + \beta \mathbf{g}) = \alpha D_{\mathbf{v}} \mathbf{f} + \beta D_{\mathbf{v}} \mathbf{g}$
3. $D_{f\mathbf{v}} \mathbf{g} = f D_{\mathbf{v}} \mathbf{g}$
4. $D_{\mathbf{v}}(f\mathbf{g}) = f D_{\mathbf{v}} \mathbf{g} + (D_{\mathbf{v}} f)\mathbf{g}$

Similarly, we invent affine connection in order to take derivative on vector field. It is a generalization of directional derivatives on manifold. An affine connection on a manifold M is a smooth map that assigns to every pair of smooth vector fields X and Y on M another smooth vector field $\nabla_X Y$ on M, satisfying the following properties:

1. $\nabla_{\alpha X_1 + \beta X_2} Y = \alpha \nabla_{X_1} Y + \beta \nabla_{X_2} Y$
2. $\nabla_X(\alpha Y_1 + \beta Y_2) = \alpha \nabla_X Y_1 + \beta \nabla_X Y_2$
3. $.\nabla_{fX}(Y) = f \nabla_X Y$
4. $\nabla_X(fY) = f \nabla_X Y + X(f) Y$

where $f_i \in C^\infty(M)$ and X_1, X_2, Y_1, Y_2 are smooth vector fields on M.
We use the Christoffel symbol to rewrite

$$\nabla_{\frac{\partial}{\partial x_i}} \frac{\partial}{\partial x_j} = \sum_{k=1}^m \Gamma_{ij}^k \frac{\partial}{\partial x_k}$$

Remark: the Christoffel symbol is different up to different basis, because Christoffel symbol is not multi-linear on functions.

Levi-Civita Connection Given a manifold, we may not be able to determine a unique affine connection. However, if we have additional two constraints, we would be able to establish a unique connection structure on the Riemannian manifolds. This is not obvious, since it is impossible to establish the Levi-Civita Connection without the Riemannian metric. The following theorem says that given the Riemannian metric, we can uniquely define an ideal affine connection, among all the possible connections, called the **Levi-Civita connection**. In the following, $\langle \cdot, \cdot \rangle$ denotes the Riemannian metric on smooth manifold M, that is, $\langle X, Y \rangle = g(X, Y)$.

Fundamental Theorem of Riemannian Geometry If (M, g) is a Riemannian manifold, then there exists a unique affine connection ∇ s.t.

(I). It is compatible with the metric, meaning that for all vector fields X, Y, Z, we have

$$X\langle Y, Z \rangle = \langle \nabla_X Y, Z \rangle + \langle Y, \nabla_X Z \rangle$$

(II). It is torsion-free, meaning that for all vector fields X, Y

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

We refer to this connection as the Levi-Civita connection.

Here, $X\langle Y, Z \rangle$ denotes the derivative of the function $\langle Y, Z \rangle$ along the vector field X, and $[X, Y]$ denotes the Lie bracket of X and Y:

$$[X, Y](f) \coloneqq X(Y(f)) - Y(X(f)) = \nabla_X \nabla_Y f - \nabla_Y \nabla_X f, \text{ where } f \in C^\infty(M)$$

Proof: Appendix 1.

It turns out that the fundamental theorem of Riemannian geometry is still doable in pseudo-Riemannian geometry, hence one can still have Levi-Civita connection. It would be interesting to explore what other results also hold true in the pseudo-Riemannian case.

1.5 Geodesics

Geodesics are curves on manifolds that generalize the idea of a straight line in a Euclidean space. One characteristic of straight lines is that they locally realize the shortest path between points, we can generalize it as g-geodesic, derived from the metric g. Another characteristic is that the tangents at all points are parallel, or, the acceleration is 0. These two characteristics are equivalent under Levi-Civita Connection, for the proof, see Appendix 2.

Geodesic as the Curve with 0 Acceleration: The smooth curve $\gamma : I \to M$ is called a geodesic, if for all $t \in I$,

$$\nabla_{\frac{d}{dt}\gamma(t)} \frac{d}{dt}\gamma(t) = 0$$

It is saying that if we take $\dot{\gamma}(t)$ as a vector field along $\gamma(t)$, then it stay constant on γ. So, the geodesics are those curves whose tangent vectors $\dot{\gamma}(t)$ are parallel transport on themselves. Here ∇ denotes the Levi-Civita Connection.

Then, take a local coordinate system on M, we can write the geodesic equation as

$$\frac{d^2}{dt^2}\gamma^k + \Gamma^k_{ij}\frac{d}{dt}\gamma^i\frac{d}{dt}\gamma^j = 0$$

This equation has a unique solution by the theory of ODE.

Geodesic as the Extremum of Length Functional Similarly, if we define the geodesics between p, q on M as the $\gamma : [a, b] \to M$, $\gamma(a) = p, \gamma(b) = q$, s.t. γ is the curve taking the minima of this functional:

$$L(\gamma) = \int_a^b \sqrt{g_{ij}(\gamma(t))(\frac{d}{dt}\gamma^i(t), \frac{d}{dt}\gamma^j(t))}dt$$

This means the length between p, q on M is minimal.

1.6 The Exponential Map

There is a natural mapping from tangent space to manifold, called exponential map. It establish an local diffeomorphism between tangent space and Riemannian manifold. In Lie algebra there is also an exponential map from TM to M, but it is different from this one in general.

Definition (Exponential map) We have $v \in T_pM$ and the geodesic $\gamma_{p,v}(t)$ defined on $[0, 1]$, then we define the $\exp_p(v)$ by

$$\exp_p(v) = \gamma_{p,v}(1)$$

It has the following properties:
(1). $\exp_p(0) = p$
(2). $d(\exp_p)_0(v) = v$

Lemma (Gauss' Lemma) Let $p \in M$, $\delta > 0$ such that $\exp_p$ is defined on $b(0, \delta)$. Let v, w be tangent vectors in T_pM with $\|v\| < \delta$. We can identify T_vT_pM with T_pM, since they are all n dimensional real vector space. Then,

$$\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\rangle_{\exp_p(v)} = \langle v, w\rangle_p$$

Proof See Appendix 3.

2 Jacobi Field

2.1 Curvature Tensor

Curvature measures the difference between a manifold with flat space. In classical differential geometry, we embed a surface into a 3-dimensional Euclidean space, and compare the difference between flat plane with the surface. In modern differential geometry, we use an intrinsic way to define the curvature.

The definition of curvature is

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

The curvature tensor measures the path dependence of parallel transport.

2.2 Definition of Jacobi Field

Definition A vector field J defined on the geodesic γ on M is called **Jacobi field** if it satisfies

$$\frac{D_\gamma^2 J}{dt^2} + R(\gamma'(t), J(t))\gamma'(t) = 0$$

Let $\Gamma(t,s) : [0,1] \times (-\varepsilon, \varepsilon)$ be a variation along the geodesic $\gamma(t)$, one can write

$$J(t) = \Gamma_{*(t,0)}(\frac{\partial}{\partial s}) \in T_{\Gamma(t,0)} M$$

One can write $S = \Gamma_* \frac{\partial}{\partial s}$, $T = \Gamma_* \frac{\partial}{\partial t}$, then

$$\nabla_T \nabla_T S = \nabla_T \nabla_S T = (R(T,S) + \nabla_{[T,S]} + \nabla_S \nabla_T)T = R(T,S)T + \nabla_S \nabla_T T = R(T,S)T$$

The first equality is because the Levi-Civita connection is torsion-free, hence

$$\nabla_T S - \nabla_S T = [T,S] = [\Gamma_* \frac{\partial}{\partial t}, \Gamma_* \frac{\partial}{\partial s}] = \Gamma_*[\frac{\partial}{\partial s}, \frac{\partial}{\partial t}] = 0$$

The second equality is by the definition of curvature is

$$R(T,S) = [\nabla_T, \nabla_S] - \nabla_{[T,S]}$$

By the theory of ODE, we know we can unique determine the Jacobi field as long as we know the initial value of $J(0)$ and $J'(0)$. Therefore, we denote $J(\gamma_v)$ to be the span of Jacobi field along γ_v, which is a $2n$-dimensional vector field.
We furthermore notice that we always have two solutions $J(t) = (\alpha + \beta t)\gamma'(t)$ to the Jacobi equation, called tangential Jacobi field, whose span is a 2-dimensional subspace of $J(\gamma)$. Let $J_0(\gamma)$ denote the $(2n-2)$-dimensional subspace of perpendicular Jacobi fields J along γ, namely, $\langle J, \gamma' \rangle = 0$.

The variation $r(t,s) : \exp_{\pi Z(s)}(tZ(s)) = \pi \circ g^t(Z(s))$, where $Z(s)$ is a C^∞ curve in the tangent bundle, such that $Z(0) = \gamma'(0)$, $DZ/ds(0) = DJ/dt(0)$, generates precisely the Jacobi field J.

2.3 Conjugate Points and Focal Points

Definition (Conjugate points). For $a \neq b$, we say that $\gamma(a)$ and $\gamma(b)$ are conjugate if there is a nonzero Jacobi field J along γ such that $J(a) = J(b) = 0$. We say that M has no conjugate points if there is no pair of conjugate points along any geodesic in M. No conjugate point is a global assumption.

If the manifold has no pair of conjugate points, then for any geodesic γ in M and nonzero Jacobi field J along γ, $J(t) = 0$ for at most one number $t \in \mathbb{R}$. Consequently, if J is perpendicular to $\gamma(t)$ at any two non-conjugate points, then they are perpendicular everywhere, since the tangential components vanish at two points.

Definition (N-Jacobi field) Let $N \subset M$ be a proper C^∞ submanifold of M, γ be a unit speed geodesic of M such that $\gamma'(0)$ is perpendicular to $T_{\gamma(0)}N$. A Jacobi field J along γ is called an N**-Jacobi field** if J can be written as

$$J(t) = \frac{\partial r}{\partial s}(t, 0)$$

$$r : (-\infty, \infty) \times (-\varepsilon, \varepsilon) \to M$$

$$(t, s) \mapsto \exp_{\pi Z(s)}(tZ(s)) = \pi \circ g^t(Z(s))$$

where $Z(s)$ is a C^∞ curve in $SN^\perp$, the unit normal bundle of N such that $Z(0) = \gamma'(0)$.

Proposition: If N is a transverse geodesic σ and $J(0) \neq 0$, then a Jacobi field J along γ is an N-Jacobi field if and only if $J(0)$ is tangent to N, J is perpendicular to γ, and $\langle J(0), J'(0)\rangle = 0$, and $\langle J(0), J'(0)\rangle = 0$.

Proof: If J is an σ-Jacobi field, $J(0) \neq 0$. $J(0)$ is tangent to σ since $J(0) = d\pi\xi$, $\xi \in T_{\dot\gamma(0)}S\sigma$. J is perpendicular to γ because $\gamma'(0) \perp T_{\gamma(0)}\sigma$ and Gauss's lemma. Then, $\langle J(0), J'(0)\rangle = \frac{d}{ds}\langle \frac{\partial r}{\partial t}, \frac{\partial r}{\partial s}\rangle = 0$.

Definition (Focal Point) For any arbitrary proper submanifold $N \subset M$, and a perpendicular unit speed geodesic γ, the point $\gamma(a)_{a\neq 0}$ is a focal point of N along γ if there is a non-trivial Jacobi field J on γ such that $J(a) = 0$. We say M have **no focal point**, if for any geodesic $N = \sigma$ has no focal point along any unit speed geodesic perpendicular to σ. The no focal point is a global assumption.

Proposition: M has no focal point if and only if for any unit speed geodesic $\gamma \in M$ and any Jacobi field J on γ such that $J(0) = 0, J'(0) \neq 0$, we have

$$\frac{d}{dt}\langle J(t), J(t)\rangle = \langle J'(t), J(t)\rangle > 0, \forall t > 0$$

Proof: Since $J(0) = 0$, the Jacobi field doesn't contain the constant term, therefore we can write it as $J(t) = J^\perp(t) + bt\gamma'(t)$. Therefore,

$$\frac{d}{dt}\langle J(t), J(t)\rangle = \frac{d}{dt}\langle J^\perp(t), J^\perp(t)\rangle + 2b^2 t$$

We only need to consider the perpendicular Jacobi field since the tangential Jacobi field always admits the proposition.

($\Rightarrow$) Suppose that M has no focal point and suppose by contradiction that $\langle J'(t_0), J(t_0)\rangle = 0$, where t_0 is the first point that $\langle J'(t_0), J(t_0)\rangle$ vanishes. We want to find a focal point on M. From the no focal point assumption, we know $J(t_0) \neq 0$. Therefore, we take the geodesic σ generated by $J(t_0)$. We see

$L(t) = J(t_0 - t)$ is a σ-Jacobi field along $\gamma(t_0 - t)$, since $\langle J'(t_0), J(t_0)\rangle = \langle L(0), L'(0)\rangle = 0$. However, $L(t_0) = J(0) = 0$, which is a focal point. Contradiction.

($\Leftarrow$) Suppose M has focal points, then there is γ and σ, $\gamma(0) \in \sigma$, $\gamma'(0) \perp \sigma$, and a non-trivial σ-Jacobi field such that $J(t_0) = 0$ for some $t_0 > 0$. Then,

$$\frac{d}{dt}\langle J(t_0 - t), J(t_0 - t)\rangle|_{t=t_0} = -2\langle J(0), J'(0)\rangle = 0$$

However, $K(t) = J(t_0 - t)$ is an N-Jacobi field with $K(0) = J(t_0) = 0$, which contradicts with the conclusion.

$\square$

Corollary: If M has no focal point and let γ be a geodesic, and $t \geq s$, then for any Jacobi field along γ with $J(0) = 0, J'(0) \neq 0$, we have

$$\|J(t)\| \geq \|J(s)\|$$

3 Geodesic Flow

Geodesic flow is a flow on tangent bundle. It send the velocity of a free particle at p to the velocity at $\gamma_v(t)$, where γ_v is the unique geodesic generated by v. This encodes the velocity information into our manifold, which is useful when we analyze the solution to geodesic equation. We will see the power of Jacobi field when inspecting geodesic flow. One can also similarly define the so-called magnetic flow, by taking Lorentz force into consideration.

3.1 Definitions and Facts

Definition (Geodesic Flow) Let $t \in \mathbb{R}$ and $v \in TM$. The geodesic flow $g^t : TM \to TM$ is a flow on TM, and is defined as

$$g^t(v) = \gamma_v'(t)$$

where $\gamma_v(t)$ is the geodesic on M parameterized by t, given the initial velocity $v \in T_{\pi v}M$. The geodesic flow leaves SM invariant. This is because $\nabla_{\dot\gamma}\langle\dot\gamma, \dot\gamma\rangle = 0$.

Definition We denote the vector field generated by the geodesic flow $X(v)$, which is a vector field of TM, namely, given $f : TM \to \mathbb{R}$, we would have

$$X(v)f = \frac{d}{dt} f(g^t v)\bigg|_{t=0}$$

We can establish an isomorphism between T_vTM and $J(\gamma_v)$. π is the natural projection from TM to M defined by $\pi : TM \to M$, $(p, v) \mapsto p$. Hence, we can define the map

$$d\pi : T_vTM \to T_{\pi v}M$$

$$(p, v, \xi) \mapsto (p, v)$$

We can also define the **connection map** by

$$K : T_vTM \to T_{\pi v}M$$

$$\xi \mapsto \frac{D_{\pi\alpha}Z}{dt}(0)$$

Where $Z : (-\epsilon, \epsilon) \to TM$, $\dfrac{dZ}{dt}(0) = \xi$, $\alpha = \pi \circ Z$. This is well-defined because connection is defined pointwise, and the covariant derivative is equivalent to Levi-Civita connection. Further more, this map is linear for every $v \in TM$ since $\nabla : \Gamma(TM) \to \Gamma(T^*M \otimes TM)$ is $\mathbb{R}$-linear.

Lemma

$$d\pi \oplus K : T_vTM \to T_{\pi v}M \oplus T_{\pi v}M$$

$$\xi \mapsto (d\pi\xi, K\xi)$$

is an isomorphism.

Proof: Indeed, they have trivial intersection, and has n dimension respectively.

Or one can have a proof by homological algebra:

$$\begin{tikzcd}
0 \arrow[r] & T_{\pi v}M \arrow[r, shift left, "\iota"] \arrow[d, leftrightarrow, "id"] & T_vTM \arrow[l, shift left, "d\pi"] \arrow[r, "K"] \arrow[d, "f"] & T_{\pi v}M \arrow[r] \arrow[d, leftrightarrow, "id"] & 0 \\
0 \arrow[r] & T_{\pi v}M \arrow[r] & T_{\pi v}M \oplus T_{\pi v}M \arrow[r] & T_{\pi v}M \arrow[r] & 0
\end{tikzcd}$$

By universal property of direct sum, we have an homomorphism from T_vTM to $T_{\pi v}M \oplus T_{\pi v}M$, and by five short lemma, we conclude that f is a linear isomorphism.

Proposition $J_\xi(t) = d\pi \circ dg^t(\xi)$ and $J'_\xi(t) = K \circ dg^t(\xi)$.

Proof: Observe that $r(t,s) = \pi \circ g^t(Z(s))$, and $J(t) = dr(\frac{\partial}{\partial s})\Big|_{s=0}$, $J'(t) = \frac{D}{dt}J(t)$.

We denote $dr(\frac{\partial}{\partial s})$ as $\frac{\partial r}{\partial s}$, $dr(\frac{\partial}{\partial t})$ as $\frac{\partial r}{\partial t}$. We should have

$$J_\xi(t) = \frac{\partial r}{\partial s}\Big|_{s=0} = \frac{\partial(\pi \circ g^t(Z(s)))}{\partial s} = d\pi \circ dg^t(\xi)$$

$$J'_\xi(t) = \frac{D}{\partial t}\frac{\partial}{\partial s}r(t,s) = \frac{D}{\partial s}\frac{\partial}{\partial t}r(t,s) = \frac{D}{\partial s}\gamma'_{Z(s)}(t) = \frac{D}{ds}g^t(Z(s)) = K \circ g^t(\xi)$$

Definition For any $v \in TM$ and $\xi \in T_vTM$, we use J_ξ to denote the unique Jacobi field on γ_v such that $(J_\xi(0), J'_\xi(0)) = (d\pi\xi, K\xi)$.

Proposition For $v \in SM$, $\xi \in T_vTM$ lies in T_vSM if and only if $\langle K \circ dg^t(\xi), g^t(v)\rangle = \langle J'_\xi(t), \gamma'_v(t)\rangle = 0$ for all $t \in \mathbb{R}$, if and only if $t \to \langle J_\xi(t), \gamma'_v(t)\rangle$ is constant.

Proof:

$$\frac{d}{dt}\langle J_\xi(t), \gamma'_v(t)\rangle = \langle J'_\xi(t), \gamma'_v(t)\rangle + \langle J'_\xi(t), \frac{D_{\gamma_v}}{dt}\gamma'_v(t)\rangle = \langle J'_\xi(t), \gamma'_v(t)\rangle = \left\langle \frac{D}{dt}\frac{\partial r}{\partial s}(t,0), \frac{\partial r}{\partial t}(t,0)\right\rangle$$

$$= \left\langle \frac{D}{ds}\frac{\partial r}{\partial t}(t,0), \frac{\partial r}{\partial t}(t,0)\right\rangle = \frac{1}{2}\frac{d}{ds}\left\langle \frac{\partial r}{\partial t}(t,s), \frac{\partial r}{\partial t}(t,s)\right\rangle\Big|_{s=0} = \frac{1}{2}\frac{d}{ds}(\|Z(s)\|)^2$$

where $Z : (-\epsilon, \epsilon) \to TM$, $\frac{dZ}{dt}(0) = \xi$. Hence, $\xi \in T_vSM$ if and only if $\|Z(s)\| \equiv 1$, if and only if $Z(s) \in SM$, if and only if $\langle J_\xi(t), \gamma'_v(t)\rangle$ is constant.

Definition (Sasaki Metric) We define the Sasaki metric on TM as the product metric on the horizontal and vertical subspace of T_vTM, namely,

$$\langle \xi, \eta\rangle_v = \langle d\pi\xi, d\pi\eta\rangle_{\pi v} + \langle K\xi, K\eta\rangle_{\pi v}$$

Proposition: $\langle \xi, X(v)\rangle = 0$ for $\xi \in T_vSM$ if and only if $\langle J_\xi(t), \gamma'_v(t)\rangle = 0$ for all $t \in \mathbb{R}$, where X is the vector field generates the geodesic flow.

Proof: $\xi \in T_vSM$, so $\langle J_\xi(t), \gamma'_v(t)\rangle$ is constant. Notice that $d\pi X(v) = v, KX(v) = 0$,

$$0 = \langle \xi, X(v)\rangle_v = \langle d\pi\xi, d\pi X(v)\rangle_{\pi v} + \langle K\xi, KX(v)\rangle_{\pi v} = \langle J(0), v\rangle = \langle J_\xi(0), \gamma'_v(0)\rangle = \langle J_\xi(t), \gamma'_v(t)\rangle$$

Corollary: If $\xi \in T_v SM$, and $\langle \xi, X(v) \rangle = 0$, then $\langle dg^t \xi, X(dg^t v) \rangle = 0$ for all $t \in \mathbb{R}$.

Proposition: Let $f : M \to N$ be a surjective local isometry, M, N are complete Riemannian manifolds, g^t are the geodesic flow, then:

(i). $df : TM \to TN$ is a surjective local isometry.

(ii). $\|dg^t \xi\| = \|dg^t d(df)\xi\|$ for all $t \in \mathbb{R}$ and $\xi \in TTM$.

(iii). $d(df)X(v) = X(dfv)$ for any $v \in TN$, where $X(v)$ is the vector field generated by the geodesic flow.

The following proposition is the bedrock of saying that the sectional curvature of a compact manifold is bounded.

Proposition: If M is compact manifold, then SM is compact.

Proof: For every x, exists a compact subset $V_x \subset U_x$ such that intV_x is open. Thus, we can find finitely many $x_1, ..., x_n$ such that $M = \bigcup_{i=1}^n V_{x_i}$. If $\pi : SM \to M$ is the projection map, then $p^{-1}(V_x) \subset p^{-1}(U_x) = U_x \times S^{n-1}$. We have $SM = \bigcup_{i=1}^n p^{-1}(\text{int } V_{x_i}) \subset \bigcup_{i=1}^n p^{-1}(V_{x_i})$, therefore, SM, as union of compact set, is compact.

Remark We describe the position of a particle subject to a force by Newton's second Law, where F, the force only depend on the position, and $F(x) = -\nabla U$ for some potential function U.

$$m\frac{d^2x}{dt^2} = F(x)$$

$$\frac{d}{dt}(mv) = -\nabla U$$

This is just the Euler-Lagrange Equation:

$$\frac{d}{dt}\frac{\partial L}{\partial v} = \frac{\partial L}{\partial x}$$

where $L : TM \to \mathbb{R}$ is given by:

$$L(x, v) = \frac{1}{2}m\langle v, v \rangle - U(x)$$

Suppose the configuration space is represented by a manifold M, and the kinetic energy is $g_x(v, v)/2$.

In the case when Lagrangian is of the form $L(x, v) = g_x(v, v)/2$, the extremum is exactly the geodesic flow. In this situation, we can use the results in geodesic flow to study the evolution of mechanical systems.

4 Appendix

4.1 Fundamental Theorem of Riemannian Geometry

If (M, g) is a Riemannian manifold, then there exists a unique affine connection ∇ such that

(I). It is compatible with the metric, meaning that for all vector fields X, Y, Z, we have

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$$

(II). It is torsion-free, meaning that for all vector fields X, Y

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

Proof We break this up into two steps.
Uniqueness: We show that if such a connection exists, then it is unique.
In the following text, $\langle \cdot, \cdot \rangle$ denotes Riemannian metric on smooth manifold M.

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle = \langle \nabla_X Y, Z\rangle + \langle [X, Z], Y\rangle + \langle \nabla_Z X, Y\rangle$$

$$Y\langle Z, X\rangle = \langle \nabla_Y Z, X\rangle + \langle Z, \nabla_Y X\rangle = \langle \nabla_Y Z, X\rangle + \langle [Y, X], Z\rangle + \langle \nabla_X Y, Z\rangle$$

$$Z\langle X, Y\rangle = \langle \nabla_Z X, Y\rangle + \langle X, \nabla_Z Y\rangle = \langle \nabla_Z X, Y\rangle + \langle [Z, Y], X\rangle + \langle \nabla_Y Z, X\rangle$$

Then

$$X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle = \langle \nabla_X Y, Z\rangle + \langle [X, Z], Y\rangle + \langle \nabla_Z X, Y\rangle +$$
$$\langle \nabla_Y Z, X\rangle + \langle [Y, X], Z\rangle + \langle \nabla_X Y, Z\rangle - (\langle \nabla_Z X, Y\rangle + \langle [Z, Y], X\rangle + \langle \nabla_Y Z, X\rangle)$$
$$= \frac{1}{2}(X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle - \langle [X, Z], Y\rangle - \langle [Y, X], Z\rangle - \langle [Y, Z], X\rangle) = \langle \nabla_X Y, Z\rangle$$

Remember that $\langle \cdot, \cdot \rangle$ is bilinear, and that Z is arbitrary, hence, this equation actually defines a mapping:

$$\langle \nabla_X Y, \cdot \rangle : TM \to \mathbb{R}$$

Then, by the properties of Riemannian metric, we uniquely define the $\nabla_X Y$.

Existence: We show that such a connection must exist.

Since we know that if such connection exists, it is unique in the form:

$$\langle \nabla_X Y, Z\rangle = \frac{1}{2}(X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle - \langle [X, Z], Y\rangle - \langle [Y, X], Z\rangle - \langle [Y, Z], X\rangle)$$

So, we just need to verify it satisfies the compatibility and torsion-free conditions.

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$$

$$\nabla_X Y - \nabla_Y X = [X, Y]$$

This is because

$$\langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle = \frac{1}{2}(X\langle Y, Z\rangle + Y\langle Z, X\rangle - Z\langle X, Y\rangle - \langle [X, Z], Y\rangle - \langle [Y, X], Z\rangle - \langle [Y, Z], X\rangle)$$

$$+\frac{1}{2}(X\langle Z,Y\rangle + Z\langle Y,X\rangle - Y\langle X,Z\rangle - \langle [X,Y],Z\rangle - \langle [Z,X],Y\rangle - \langle [Z,Y],X\rangle) = X\langle Y,Z\rangle$$

This proves the compatibility.
also,

$$\langle \nabla_X Y, Z\rangle - \langle \nabla_Y X, Z\rangle = \langle \nabla_X Y - \nabla_Y X, Z\rangle =$$

$$\frac{1}{2}(X\langle Y,Z\rangle + Y\langle Z,X\rangle - Z\langle X,Y\rangle - \langle [X,Z],Y\rangle - \langle [Y,X],Z\rangle - \langle [Y,Z],X\rangle) -$$

$$\frac{1}{2}(Y\langle X,Z\rangle + X\langle Z,Y\rangle - Z\langle Y,X\rangle - \langle [Y,Z],X\rangle - \langle [X,Y],Z\rangle - \langle [X,Z],Y\rangle) = \langle [X,Y],Z\rangle$$

Hence,

$$\langle \nabla_X Y - \nabla_Y X - [X,Y], Z\rangle \equiv 0$$

implies

$$\nabla_X Y - \nabla_Y X = [X,Y]$$

This proves the torsion-free.

4.2 The equivalence of two formulations of geodesic

The smooth curve $\gamma : I \to M$ is called a geodesic, if for all $t \in I$,

$$\nabla_{\frac{d}{dt}\gamma(t)} \frac{d}{dt}\gamma(t) = 0$$

Another definition of geodesic is the curve $\gamma : [a, b] \to M$, $\gamma(a) = p, \gamma(b) = q$, such that γ takes the minimum of this functional:

$$L(\gamma) = \int_a^b \sqrt{g_{ij}(\gamma(t))(\frac{d}{dt}\gamma^i(t), \frac{d}{dt}\gamma^j(t))}dt$$

This means the length between p, q on M is minimal.
The solution is equivalent to:

$$E(\gamma) = \int_a^b g_{ij}(\gamma(t))(\frac{d}{dt}\gamma^i(t), \frac{d}{dt}\gamma^j(t))dt$$

this is by Cauchy-Schwarz inequality, since we can take this functional as an inner product, then

$$L^2 \leq (b - a)E$$

So, the minimality of E implies the minimality L.
Then, we plug in the Euler-Lagrange equation, where the Lagrangian is:

$$E = g_{ij}(\gamma(t))(\frac{d}{dt}\gamma^i(t), \frac{d}{dt}\gamma^j(t))$$

The Euler-Lagrange Equation is:

$$\frac{\partial E}{\partial x^k} = \frac{d}{dt}\frac{\partial E}{\partial \dot{x^i}}$$

$$\frac{\partial E}{\partial x^k} = \frac{\partial g_{ij}}{\partial x^k}\dot{\gamma}^i\dot{\gamma}^j$$

$$\frac{d}{dt}\frac{\partial E}{\partial \dot{x^k}} = \frac{d}{dt}[g_{ik}\dot{\gamma}^i + g_{kj}\dot{\gamma}^j] = \frac{\partial g_{ik}}{\partial x^j}\dot{\gamma}^j\dot{\gamma}^i + g_{ik}\ddot{\gamma}^i + \frac{\partial g_{kj}}{\partial x^i}\dot{\gamma}^i\dot{\gamma}^j + g_{kj}\ddot{\gamma}^j$$

So, equating both sides:

$$\frac{\partial g_{ik}}{\partial x^j}\dot{\gamma}^j\dot{\gamma}^i + g_{ik}\ddot{\gamma}^i + \frac{\partial g_{kj}}{\partial x^i}\dot{\gamma}^i\dot{\gamma}^j + g_{kj}\ddot{\gamma}^j = \frac{\partial g_{ij}}{\partial x^k}\dot{\gamma}^i\dot{\gamma}^j$$

$$(\frac{\partial g_{ik}}{\partial x^j} + \frac{\partial g_{jk}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^k})\dot{\gamma}^i\dot{\gamma}^j + 2g_{lk}\ddot{\gamma}^l = 0$$

Also, in Levi-Civita connection,

$$\Gamma^k_{ij} = \frac{1}{2}(\frac{\partial g_{il}}{\partial x^j} + \frac{\partial g_{jl}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^k})g^{lk}$$

We should have:

$$\frac{d^2}{dt^2}\gamma^k + \Gamma^k_{ij}\frac{d}{dt}\gamma^i\frac{d}{dt}\gamma^j = 0$$

Since the solution to ODE is unique, two definitions of geodesics coincide.

$$\nabla_{\frac{d}{dt}\gamma(t)} \frac{d}{dt}\gamma(t) = 0$$

Hence, the Levi-Civita connection defines the parallel transport on the geodesic.

4.3 Gauss' Lemma

Let $p \in M$, $\delta > 0$ such that $\exp_p$ is defined on $b(0, \delta)$. Let v, w be tangent vectors in T_pM with $\|v\| < \delta$. We can identify T_vT_pM with T_pM, since they are all n dimensional real vector space. Then,

$$\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\rangle_{\exp_p(v)} = \langle v, w\rangle_p$$

Proof: We write

$$\Gamma : [0,1] \times (-\epsilon, \epsilon) \to M$$
$$(t, s) \mapsto \exp_p(t(v + sw))$$

Therefore, $\Gamma(t, s)$ is a surface embedded in M.

$$\left.\frac{\partial \Gamma}{\partial t}\right|_{s=0,t=1} = d(\exp_p)_{tv}(v)$$

$$\left.\frac{\partial \Gamma}{\partial s}\right|_{s=0,t=1} = d(\exp_p)_{tv}(w)$$

$$\left.\frac{\partial \Gamma}{\partial t}\right|_{s=0,t=0} = \left.\frac{\partial \Gamma}{\partial s}\right|_{s=0,t=0} = 0$$

Therefore,

$$\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\rangle_{\exp_p(v)} = \langle \left.\frac{\partial \Gamma}{\partial t}\right|_{s=0,t=1}, \left.\frac{\partial \Gamma}{\partial s}\right|_{s=0,t=1} \rangle$$

$$\langle v, w\rangle_p = \langle \left.\frac{\partial \Gamma}{\partial t}\right|_{s=0,t=0}, \left.\frac{\partial \Gamma}{\partial s}\right|_{s=0,t=0} \rangle$$

However,

$$\frac{\partial}{\partial t}\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial s}\rangle = \langle \frac{D}{dt}\frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial s}\rangle + \langle \frac{D}{dt}\frac{\partial \Gamma}{\partial s}, \frac{\partial \Gamma}{\partial t}\rangle = \langle \nabla_{\dot\gamma}\dot\gamma, \frac{\partial \Gamma}{\partial s}\rangle + \langle \frac{D}{dt}\frac{\partial \Gamma}{\partial s}, \frac{\partial \Gamma}{\partial t}\rangle = \langle \frac{D}{dt}\frac{\partial \Gamma}{\partial s}, \frac{\partial \Gamma}{\partial t}\rangle = \langle \frac{D}{ds}\frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial t}\rangle$$

$$= \frac{1}{2}\frac{\partial}{\partial s}\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial t}\rangle = \frac{1}{2}\frac{\partial}{\partial s}\langle v + sw, v + sw\rangle = \langle v + sw, w\rangle$$

When $s = 0$,

$$\langle v + sw, w\rangle = 0$$

So,

$$\frac{\partial}{\partial t}\langle \frac{\partial \Gamma}{\partial t}, \frac{\partial \Gamma}{\partial s}\rangle$$

is constant.

Therefore, we have

$$\langle d(\exp_p)_v(v), d(\exp_p)_v(w)\rangle_{\exp_p(v)} = \langle v, w\rangle_p$$

5 Reference

[1] Burns, Keith, Gidea, Marian, CRC Press, Differential Geometry and Topology With a View to Dynamical Systems

[2] Lee, John M., Springer, Introduction to Smooth Manifolds

[3] Hao-Tong Yan, Anosov Geodesic Flows on Surfaces